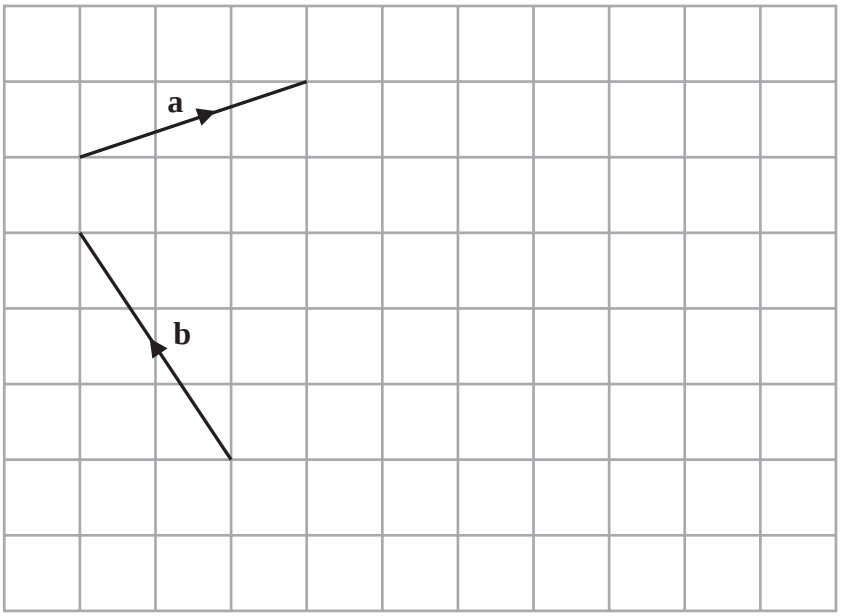
10 The vector **a** and the vector **b** are shown on the grid.



(a) On the grid, draw and label the vector **a** – **b**

(1)

p and **q** are vectors such that

$$\mathbf{p} = \begin{pmatrix} 3 \\ c \end{pmatrix} \text{ and } \mathbf{q} = \begin{pmatrix} d \\ -2 \end{pmatrix}$$

Given that $2\mathbf{p} - 3\mathbf{q} = \begin{pmatrix} -9 \\ 14 \end{pmatrix}$

(b) find the value of c and the value of d.

$c =$ .....

$d =$ .....

(3)

(Total for Question 10 is 4 marks)